

BIT Sindri
Department of Chemistry
B. Tech 2nd Semester/
Engineering Chemistry

Unit3

- Q1. Explain principle, construction and advantages of Lithium ion battery.
- Q2. Define Green Hydrogen Technology. How does it work? Explain with labelled diagram.
- Q3. Distinguish between Lithium ion battery and sodium ion battery. Write at least 7 points.
- Q4. Discuss, in detail, Hydrogen Oxygen fuel cell. Propose two advantages and two disadvantages of fuel cell.
- Q5. List various types of solar cells. Compare and contrast Monocrystalline solar panels and polycrystalline solar panels.
- Q6. How does a fuel cell differ from traditional methods of energy generation? How is a fuel cell different than a battery?
- Q7. How Comparable are Sodium-Ion Batteries to Lithium-Ion Counterparts? Interpret with suitable justification.
- Q8. Write short note on following
(I) Emerging trends of fuel cell (II) Fibre cell (III) Green chemistry
(IV) amorphous silicon solar cell
- Q9. Elaborate various applications of Lithium ion battery in detail.
- Q10. Explain working principle and advantages of Sodium ion battery.
- Q11. Outline various Principles of Green Chemistry. Illustrate any three in detail.
- Q12. Write short notes on following
(I) Proton exchange membrane (II) Solid oxide fuel cell
(III) Alkali fuel cell (IV) Drawbacks of fuel cells
- Q13. Classify different types of Electrolyzers in green hydrogen technology. Why green hydrogen is needed?
- Q14. Construct a simple solar cell. Identify different materials used in construction of solar cell. Analyze In brief working Principle of solar cell.

Unit4

Q1. Define Nanomaterials. How can you classify Nano materials into different categories depending on the number of dimensions .What are the aims and objectives of Nano science.

Q2. Compare and contrast Top down approach and Bottom up approach in the nano Technology with one example of each. Write at least 7 points.

Q3. List different materials used in super capacitor. Explain the classification of super capacitor with the help of a flowchart.

Q4. Discuss advantages and disadvantages of super capacitor in detail.

Q5. Compile applications of Nanomaterials in various Engineering field.

Q6. Justify the statement “Nano technology is one of the remarkable and indispensable area in the fields of industry and research. “ by giving some suitable examples.

Q7. Do you agree with the statement that “There is a need to develop devices that are cheaper harmless renewable and energy efficient.” How green nanotechnology can help in this?

Q8. Define Green nanotechnology. Write various applications of green Nanomaterials in electronic and electrical industries.

Q9. Write short notes on

(I)ball milling synthesis method (II)synthesis of a super capacitor

(III)Green Nanotechnology. (IV)Quantum dots

Q10. Define super capacitor. Construct a simplified and well labelled schematic illustration of a super capacitor. What are the differences between super capacitor and a battery?

Q11. Elaborate Aim of Green nanotechnology.

Propose practical solutions to maximize the techniques of Green Nanotechnology. (3+4)

Q12. Discuss RF Sputtering deposition method to synthesize Nanomaterials. Write innovations and advancements in RF Sputtering technology. Give some applications of this method.

Q13. Discuss in detail

(I) Pulsed laser deposition method. (II) Thin film deposition method.

Q14. Write short notes on

(I) Synthesis of Green Nanomaterials. (II)Nanoscale phenomenon

(III)Supercapacitor (IV)RF Sputtering synthesis method

Q15. Explain, in detail, any one Top down and any one Bottom up approach for Nanomaterials synthesis.